$$\mathbb{P}(\overset{g}{X} \geq t) \leq \frac{\mathbb{E}(X)}{t}$$

$X \geq 0$

g increasing

$$X \geq t \iff g(X) \geq g(t)$$

$$g(x) = X^2$$

$$\mathbb{P}(\underset{(X-\mu_X)^2}{X^2} \geq \underbrace{t^2}_{s}) \leq \frac{\mathbb{E}(\overset{Var(X)}{X^2})}{\underbrace{t^2}_{s}}$$

$$g(x) = e^{\lambda X} \qquad \lambda \geq 0 \qquad 10$$

$$\mathbb{P}(e^{\lambda(X-\mu)} \geq \underbrace{e^{\lambda t}}_{t}) \leq \frac{\mathbb{E}(e^{\lambda(X-\mu)})}{\underbrace{e^{\lambda t}}_{t}} \leq \cdots$$

$\inf_{\lambda \geq 0}$

$$X \sim \mathcal{N}(0, \sigma^2)$$

$$\mathbb{E}(e^{\lambda X}) = \int e^{\lambda x} \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\frac{x^2}{\sigma^2}} dx$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma} \int e^{-\frac{1}{2} x^2/\sigma^2 + \lambda x} dx$$

$$\rightsquigarrow -\frac{(x-a)^2}{2\sigma^2} = -\frac{x^2}{2\sigma^2} - \frac{a^2}{2\sigma^2} + 2ax/2\sigma^2$$

$b = \sigma$

$$\lambda = \frac{a}{b^2} \Rightarrow a = \lambda\sigma^2$$

$$\Rightarrow \mathbb{E}\left(e^{\lambda X}\right) \overset{\leq}{=} \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{\left\{-\frac{(x-a)^2}{2b^2} + \frac{a^2}{2b^2}\right\}} dx$$

$$= e^{a^2/2b^2} = e^{\lambda^2\sigma^4/2\sigma^2} = \boxed{e^{\frac{\lambda^2\sigma^2}{2}}}$$

کران بالای احتمال $= \inf_{\lambda \geq 0} \underbrace{e^{\frac{\lambda^2\sigma^2}{2}} \cdot e^{-\lambda t}}$

$$\frac{\lambda^2\sigma^2}{2} - \lambda t \xrightarrow{\partial/\partial\lambda} \lambda\sigma^2 - t = 0 \Rightarrow \lambda = t/\sigma^2 \geq 0$$

$$e^{\frac{t^2}{2\sigma^4}\cdot\frac{\sigma^2}{2}} \cdot e^{-t^2/\sigma^2} = e^{\frac{-t^2}{2\sigma^2}}$$

$X \sim N(\mu, \sigma^2)$

$$\mathbb{P}(X - \mu \geq t) \leq e^{-t^2/2\sigma^2}$$

Distribution-Free

A family of SubGaussian distribution

$$\in f_X \Leftrightarrow \exists \sigma \geq 0 : \mathbb{E}\left(e^{\lambda X}\right) \leq e^{\lambda^2\sigma^2/2}$$

$$\begin{cases} \mathbb{P}(X = 1) = 0.5 \\ \mathbb{P}(X = -1) = 0.5 \end{cases}$$

$$\mathbb{E}(e^{\lambda X}) = \underbrace{\frac{1}{2}e^{\lambda} + \frac{1}{2}e^{-\lambda}}_{} \overset{?}{\leq} e^{\lambda^2\sigma^2/2}$$

$$e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

$$\sum_{k=0}^{\infty} \frac{\lambda^{2k}}{(2k)!} = (\lambda^2)^k$$

$$\leq 1 + \sum_{k=1}^{\infty} \frac{(\lambda^2/2)^k}{k!} = e^{\lambda^2/2}$$

$$2k! = \underbrace{1 \times 2 \times \cdots \times k}_{k!} \times \underbrace{k+1}_{\geq 2} \times \cdots \times \underbrace{2k}_{\geq 2}$$

$\Rightarrow$ SubG $\sigma^2 = 1$

$X \in [a,b]$, $\underline{\mathbb{E}(X)=0} \Rightarrow X$ is subG

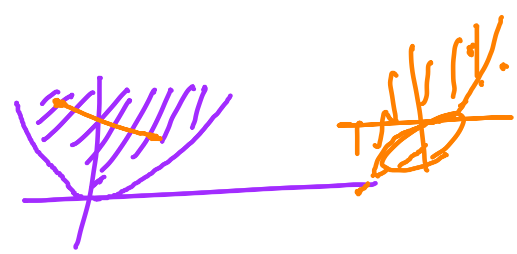
$$\mathbb{E}(e^{\lambda X}) = \mathbb{E}(e^{\lambda(X - \mathbb{E}(X))})$$

$X' \perp\!\!\!\perp X$

$X' \sim f_X$

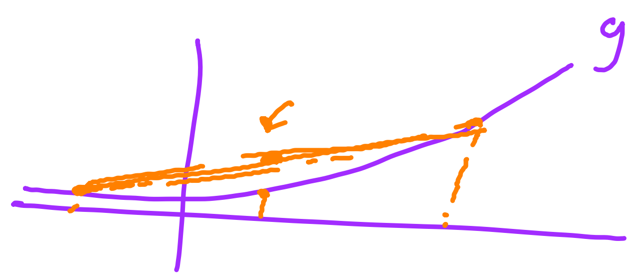
$g(X')$

$$= \mathbb{E}(\underline{e^{\lambda(X - \mathbb{E}(X'))}})$$



g is a convex function

$$g(\mathbb{E}(X)) \leq \mathbb{E}(g(X))$$



$$\mathbb{E}(e^{\lambda X} \cdot e^{-\lambda \mathbb{E}(X')})$$

$$\leq \mathbb{E}_{X,X',\epsilon}(e^{\lambda\epsilon(X - X')})$$

$$\underbrace{\tfrac{1}{2}}_{P(\epsilon=1)} \mathbb{E}\left(e^{\lambda(x-x')} \,\middle|\, \epsilon = 1 \right)$$

$$+$$

$$\underbrace{\tfrac{1}{2}}_{P(\epsilon=-1)} \mathbb{E}\left(e^{\lambda(x'-x)} \,\middle|\, \epsilon = -1 \right)$$

$$= \mathbb{E}_{x, x', \epsilon}\left(e^{\lambda \epsilon (x - x')} \right)$$

$$\mathbb{E}_{X,Y}\left(g(X,Y) \right) = \mathbb{E}_X\left(\mathbb{E}_Y\left(g(X,Y) / X \right) \right)$$

[illegible]

$$= \mathbb{E}_{x, x', \epsilon}\left(e^{\lambda \epsilon (x - x')} \right)$$

$$= \mathbb{E}_{x, x'} \underbrace{\mathbb{E}_{\epsilon \mid x, x'} \left(e^{\lambda (x - x') \epsilon} \right)}_{\leq e^{\lambda^2 (x - x')^2 / 2}}$$

$$\leq \mathbb{E}_{x, x'} \underbrace{e^{\lambda^2 (x - x')^2 / 2}}_{\leq e^{\lambda^2 (b-a)^2 / 2}}$$

a $\quad$ b

$$\leq \underbrace{e^{\lambda^2 (b-a)^2 / 2}}_{e^{\lambda^2 \sigma^2 / 2}} \qquad \sigma = b - a$$

$$\begin{cases} X_1 & \text{subG} & \sigma_1 & \text{indep.} \\ X_2 & \text{subG} & \sigma_2 & \\ X_1 + X_2 & \text{subG} & \sqrt{\sigma_1^2 + \sigma_2^2} & \end{cases}$$